

**Experiment 1:** Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

### 1. student placement prediction dataset

| CGPA | Aptitude Score | Communication Score | Placement  |
|------|----------------|---------------------|------------|
| 9.2  | 90             | 88                  | Placed     |
| 8.8  | 85             | 82                  | Placed     |
| 8.5  | 80             | 79                  | Placed     |
| 7.2  | 72             | 70                  | Not Placed |
| 6.8  | 68             | 65                  | Not Placed |
| 9    | 92             | 90                  | Placed     |
| 7.5  | 75             | 74                  | Not Placed |
| 8.6  | 84             | 81                  | Placed     |
| 6.5  | 60             | 58                  | Not Placed |
| 8.9  | 88             | 86                  | Placed     |

#### CODE:

```
import pandas as pd

df = pd.read_csv("student placement prediction .csv")

concepts = df.iloc[:, :-1].values

target = df.iloc[:, -1].values

hypothesis = ['0'] * len(concepts[0])

print("\nFIND-S Training Process:\n")

for i in range(len(target)):

    if target[i] == "Yes" or target[i] == "Placed":
```

```

if hypothesis == ['0'] * len(concepts[0]):
    hypothesis = list(concepts[i])

else:
    for j in range(len(hypothesis)):
        if hypothesis[j] != concepts[i][j]:
            hypothesis[j] = '?'

print("Step", i + 1, ":", hypothesis)

print("\nFinal Most Specific Hypothesis:")
print(hypothesis)

```

OUTPUT:

```

...
FIND-S Training Process:

Step 1 : [np.float64(9.2), np.float64(90.0), np.float64(88.0)]
Step 2 : ['?', '?', '?']
Step 3 : ['?', '?', '?']
Step 6 : ['?', '?', '?']
Step 8 : ['?', '?', '?']
Step 10 : ['?', '?', '?']

Final Most Specific Hypothesis:
['?', '?', '?']

```

## 2. loan approval prediction dataset

| Income (₹ Lakhs) | Credit Score | Loan Amount (₹ Lakhs) | Loan Approved |
|------------------|--------------|-----------------------|---------------|
| 8                | 780          | 5                     | Yes           |
| 7                | 760          | 4                     | Yes           |
| 6                | 720          | 6                     | Yes           |
| 4                | 620          | 8                     | No            |
| 3                | 580          | 10                    | No            |
| 9                | 800          | 4                     | Yes           |
| 5                | 650          | 7                     | No            |
| 8                | 770          | 5                     | Yes           |
| 4                | 600          | 9                     | No            |
| 7                | 740          | 6                     | Yes           |

### Code:

```
import pandas as pd

df = pd.read_csv("loan approval prediction .csv")

concepts = df.iloc[:, :-1].values

target = df.iloc[:, -1].values

hypothesis = None

print("\nTraining Process:\n")

for i in range(len(concepts)):

    if target[i] == "Yes":

        if hypothesis is None:

            hypothesis = concepts[i].astype(str)

        else:

            for j in range(len(hypothesis)):

                if str(hypothesis[j]) != str(concepts[i][j]):
```

```

        hypothesis[j] = "?"

    print(f'Step {i+1}: {hypothesis}')

print("\nFinal Hypothesis:")

print(hypothesis)

```

### Output:

```

...
    Training Process:

    Step 1: ['8' '780' '5']
    Step 2: ['?' '?' '?']
    Step 3: ['?' '?' '?']
    Step 6: ['?' '?' '?']
    Step 8: ['?' '?' '?']
    Step 10: ['?' '?' '?']

    Final Hypothesis:
    ['?' '?' '?']

```

### 3. disease diagnosis

| Temperature (°C) | Heart Rate (bpm) | Oxygen Level (%) | Disease  |
|------------------|------------------|------------------|----------|
| 39               | 110              | 94               | Positive |
| 38.5             | 108              | 95               | Positive |
| 37               | 78               | 99               | Negative |
| 39.2             | 115              | 93               | Positive |
| 36.8             | 75               | 98               | Negative |
| 38.8             | 112              | 94               | Positive |
| 37.2             | 80               | 98               | Negative |

### Code:

```

import pandas as pd

df = pd.read_csv("disease diagnosis .csv") # Corrected filename here

```

```

attributes = df.columns[:-1]
hypothesis = None

for _, row in df.iterrows():
    if row["Disease"] == "Positive":
        if hypothesis is None:
            hypothesis = list(row[attributes])
        else:
            for i in range(len(attributes)):
                if hypothesis[i] != row[attributes[i]]:
                    hypothesis[i] = "?"

print("Most Specific Hypothesis:")
print(hypothesis)

```

Output:

```

... Most Specific Hypothesis:
   ['?', '?', '?']

```

#### 4. Employee Promotion Prediction dataset

| Experience (Years) | Performance Score | Training Hours | Promotion |
|--------------------|-------------------|----------------|-----------|
| 10                 | 95                | 50             | Yes       |
| 8                  | 90                | 45             | Yes       |
| 7                  | 88                | 40             | Yes       |
| 3                  | 65                | 20             | No        |
| 2                  | 60                | 18             | No        |
| 9                  | 93                | 48             | Yes       |
| 4                  | 70                | 25             | No        |
| 8                  | 89                | 42             | Yes       |
| 2                  | 58                | 15             | No        |
| 6                  | 85                | 38             | Yes       |

Code:

```
import pandas as pd

data = pd.read_csv("employee_promotion_prediction.csv")

hypothesis = None

attributes = list(data.columns[:-1])

for index, row in data.iterrows():

    if row["Promotion"] == "Yes":

        if hypothesis is None:

            hypothesis = list(row[attributes])

        else:

            for i in range(len(attributes)):

                if hypothesis[i] != row[attributes[i]]:

                    hypothesis[i] = "?"

print("\nMost Specific Hypothesis:")

for attr, value in zip(attributes, hypothesis):
```

```
print(f'{attr}: {value}')
```

### Output:

```
... Most Specific Hypothesis:  
['?', '?', '?']
```

### 5. Fruit Classification dataset

| Weight (g) | Diameter (cm) | Sweetness (%) | Fruit  |
|------------|---------------|---------------|--------|
| 150        | 7.5           | 90            | Apple  |
| 160        | 7.8           | 88            | Apple  |
| 140        | 7.2           | 91            | Apple  |
| 120        | 5.5           | 70            | Orange |
| 125        | 5.8           | 72            | Orange |
| 130        | 6             | 74            | Orange |
| 155        | 7.6           | 89            | Apple  |
| 118        | 5.4           | 69            | Orange |
| 148        | 7.4           | 92            | Apple  |
| 122        | 5.7           | 71            | Orange |

### Code:

```
import pandas as pd  
data = pd.read_csv("fruit_classification.csv")  
print("Fruit Classification Dataset:\n")  
print(data)  
hypothesis = None
```

```

print("\nSteps of FIND-S Algorithm:\n")
for index, row in data.iterrows():
    if row["Fruit"] == "Apple":
        if hypothesis is None:
            hypothesis = list(row[:-1])
        else:
            for i in range(len(hypothesis)):
                if hypothesis[i] != row[i]:
                    hypothesis[i] = "?"

    print(f'After Positive Example {index + 1}:')
    print(hypothesis)
    print()

print("Final Specific Hypothesis:")
print(hypothesis)

```

### Output:

```

... Most Specific Hypothesis:
    ['?', '?', '?']

```